IoE and AI driven systems
LAB OUTPUTS

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EXPERIMENT NO:5

A.Simulation and Performance Evaluation of RFID/NFC-Based Device Identification and IoE Data Integration using Node-RED, MQTT.

PROGRAM CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(24)

# ============================================================
# EXPERIMENT A: RFID/NFC DEVICE IDENTIFICATION
# ============================================================

devices = [
    ("RFID001", "RFID"),
    ("RFID002", "RFID"),
    ("NFC003", "NFC"),
    ("RFID004", "RFID"),
    ("NFC005", "NFC"),
    ("RFID006", "RFID"),
    ("NFC007", "NFC"),
    ("RFID008", "RFID"),
    ("NFC009", "NFC"),
    ("RFID010", "RFID")
]

results = []

print("=" * 70)
print("EXPERIMENT A: RFID/NFC DEVICE IDENTIFICATION")
print("=" * 70)

# 1. Device identification
for device_id, technology in devices:
```

```
    success_probability = 0.96 if technology == "RFID" else 0.98

    identified = np.random.random() < success_probability

    delay = (
        np.random.normal(35, 3)
        if technology == "RFID"
        else np.random.normal(25, 2)
    )

    message_size = 256 if technology == "RFID" else 192

    results.append([
        device_id,
        technology,
        "Success" if identified else "Failed",
        round(max(delay, 1), 2),
        message_size
    ])

df = pd.DataFrame(
    results,
    columns=[
        "Device ID",
        "Technology",
        "Identification Status",
        "Delay (ms)",
        "Message Size (bytes)"
    ]
)

identified_devices = df[
    df["Identification Status"] == "Success"
]

# 2. Simulate MQTT data integration
mqtt_sent = len(identified_devices)
mqtt_delivery_rate = 97.0

mqtt_received = round(
    mqtt_sent * mqtt_delivery_rate / 100
)
```

```
# 3. Simulate cloud dashboard integration
cloud_received = round(mqtt_received * 0.98)

print("\nDEVICE IDENTIFICATION RESULTS")
print("-" * 70)
print(df.to_string(index=False))

print("\nMQTT DATA INTEGRATION")
print("-" * 70)
print("Devices detected       :", len(df))
print("Successfully identified:", len(identified_devices))
print("MQTT messages sent     :", mqtt_sent)
print("MQTT messages received :", mqtt_received)
print("Cloud records received :", cloud_received)
print("Modelled MQTT delivery :",
      round(mqtt_received / mqtt_sent * 100, 2)
      if mqtt_sent else 0, "%")

# 4. RFID vs NFC comparison
comparison = df.groupby("Technology").agg(
    Devices=("Device ID", "count"),
    Average_Delay=("Delay (ms)", "mean")
).reset_index()

success = df.groupby("Technology").apply(
    lambda x: (x["Identification Status"] == "Success").mean() * 100,
    include_groups=False
).reset_index(name="Success_Rate")

comparison = comparison.merge(success, on="Technology")

print("\nTECHNOLOGY COMPARISON")
print("-" * 70)
print(comparison.round(2).to_string(index=False))

# 5. Graph: Identification status
plt.figure(figsize=(7, 4))
df["Identification Status"].value_counts().plot(kind="bar")
plt.title("RFID/NFC Device Identification Status")
plt.xlabel("Status")
plt.ylabel("Number of Devices")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```

```
# 6. Graph: RFID vs NFC success rate
plt.figure(figsize=(7, 4))
plt.bar(comparison["Technology"], comparison["Success_Rate"])
plt.title("RFID vs NFC Identification Success")
plt.xlabel("Technology")
plt.ylabel("Observed Success Rate (%)")
plt.ylim(0, 105)
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

# 7. Graph: Device identification delay
plt.figure(figsize=(9, 4))
plt.bar(df["Device ID"], df["Delay (ms)"])
plt.title("Device Identification Delay")
plt.xlabel("Device ID")
plt.ylabel("Delay (ms)")
plt.xticks(rotation=45)
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

# 8. Graph: Data integration flow
plt.figure(figsize=(8, 4))
stages = ["Identified", "MQTT Received", "Cloud Received"]
counts = [len(identified_devices), mqtt_received, cloud_received]
plt.bar(stages, counts)
plt.title("RFID/NFC to MQTT and Cloud Data Flow")
plt.xlabel("Integration Stage")
plt.ylabel("Number of Records")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

print("\nCONCLUSION")
print("-" * 70)
print("RFID/NFC tags are modelled for device identification.")
print("Identified device data is modelled as MQTT messages.")
print("Received messages are modelled as cloud dashboard records.")
print("Experiment A completed.")
```

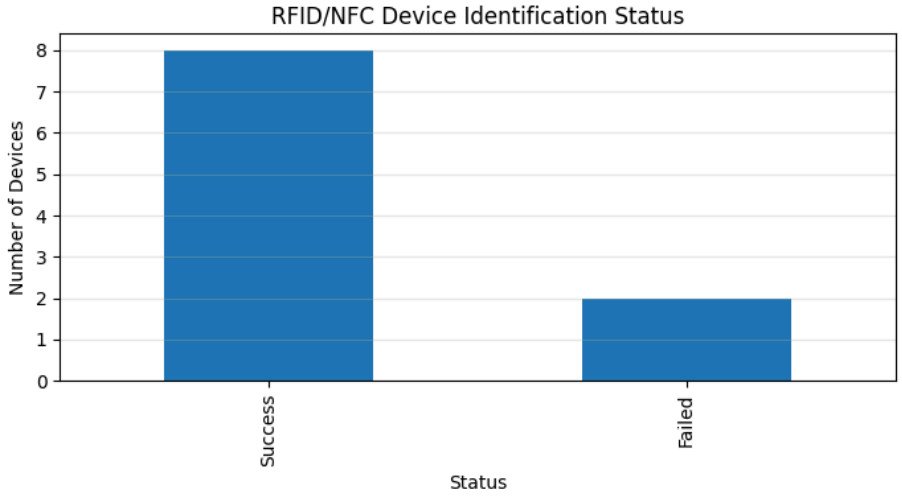
OUTPUTS:

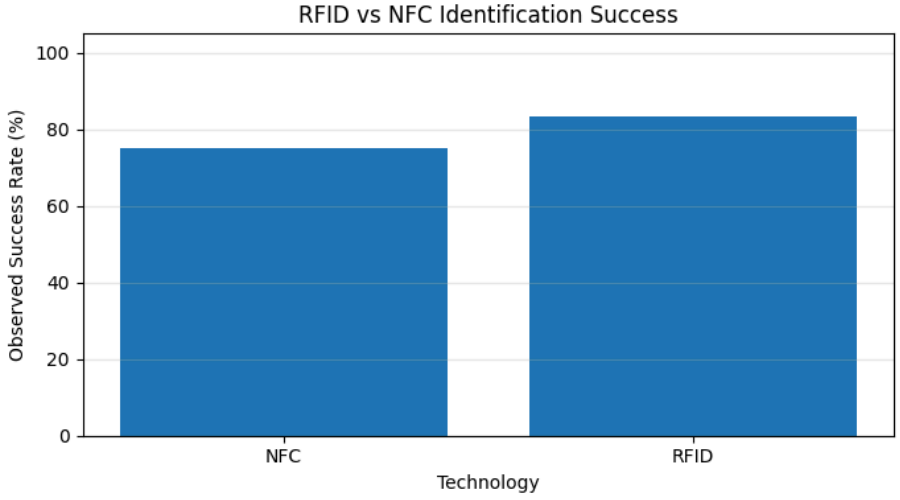
```
======================================================================
EXPERIMENT A: RFID/NFC DEVICE IDENTIFICATION
======================================================================

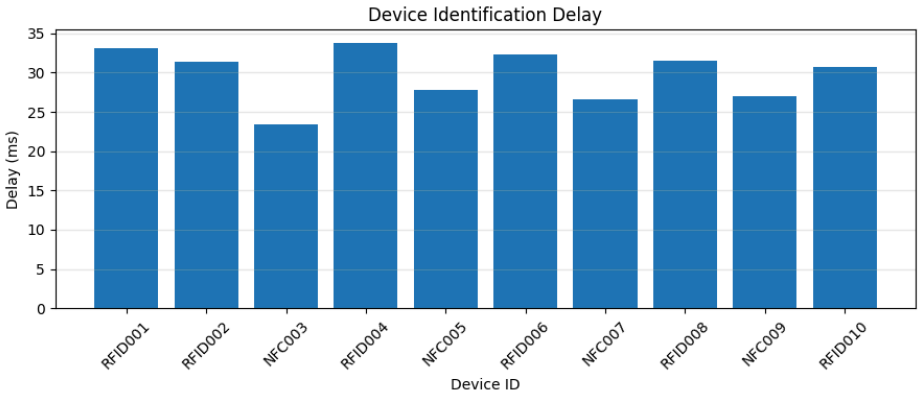
DEVICE IDENTIFICATION RESULTS
----------------------------------------------------------------------
Device ID Technology Identification Status  Delay (ms)  Message Size (bytes)
  RFID001       RFID                Failed       33.17                  256
  RFID002       RFID               Success       31.32                  256
   NFC003        NFC                Failed       23.38                  192
  RFID004       RFID               Success       33.77                  256
   NFC005        NFC               Success       27.82                  192
  RFID006       RFID               Success       32.30                  256
   NFC007        NFC               Success       26.58                  192
  RFID008       RFID               Success       31.52                  256
   NFC009        NFC               Success       26.95                  192
  RFID010       RFID               Success       30.67                  256

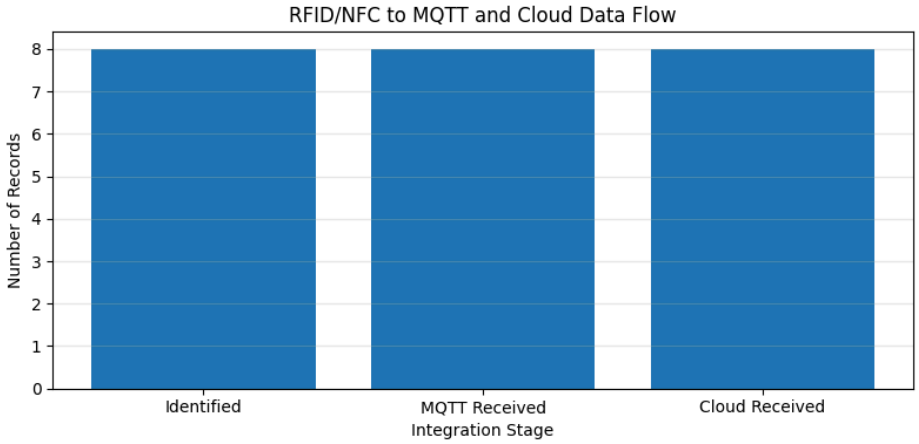
MQTT DATA INTEGRATION
----------------------------------------------------------------------
Devices detected       : 10
Successfully identified: 8
MQTT messages sent     : 8
MQTT messages received : 8
Cloud records received : 8
Modelled MQTT delivery : 100.0 %

TECHNOLOGY COMPARISON
----------------------------------------------------------------------
Technology  Devices  Average_Delay  Success_Rate
       NFC        4          26.18         75.00
      RFID        6          32.12         83.33
```









CONCLUSION

RFID/NFC tags are modelled for device identification.
Identified device data is modelled as MQTT messages.
Received messages are modelled as cloud dashboard records.
Experiment A completed.

B.Modelling and Simulation of a 4-Layer IoE Architecture (Perception–Network–Middleware–Application) using Node-RED, MQTT and a Cloud Dashboard

PROGRAM:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(42)

# ============================================================
# EXPERIMENT B: FOUR-LAYER IoE ARCHITECTURE
# ============================================================
```

```
print("=" * 75)
print("EXPERIMENT B: FOUR-LAYER IoE ARCHITECTURE SIMULATION")
print("=" * 75)

# Generate sensor/device data
number_of_devices = 20

sensor_data = np.random.uniform(20, 40, number_of_devices)

df = pd.DataFrame({
    "Device ID": [f"DEVICE_{i+1:02d}"
                  for i in range(number_of_devices)],
    "Sensor Value": np.round(sensor_data, 2),
    "Unit": ["°C"] * number_of_devices
})

# Four architecture layers
layers = [
    "Perception",
    "Network",
    "Middleware",
    "Application"
]

# Percentage of messages processed successfully per layer
success_rates = [98, 97, 99, 98]

# Simulate messages passing through each layer
input_messages = number_of_devices
layer_results = []

for layer, rate in zip(layers, success_rates):

    processed = round(input_messages * rate / 100)

    layer_results.append([
        layer,
        input_messages,
        processed,
        rate
    ])

    input_messages = processed
```

```
layer_df = pd.DataFrame(
    layer_results,
    columns=[
        "IoE Layer",
        "Input Messages",
        "Processed Messages",
        "Configured Success Rate (%)"
    ]
)

# MQTT messaging model
mqtt_sent = layer_df.iloc[1]["Processed Messages"]
mqtt_received = round(mqtt_sent * 0.97)

# Application/cloud dashboard
cloud_received = round(mqtt_received * 0.98)

dashboard = pd.DataFrame({
    "Dashboard Metric": [
        "Connected Devices",
        "Sensor Messages Generated",
        "MQTT Messages Sent",
        "MQTT Messages Received",
        "Cloud Records Received",
        "Average Sensor Value (°C)",
        "Maximum Sensor Value (°C)",
        "Minimum Sensor Value (°C)"
    ],
    "Value": [
        number_of_devices,
        number_of_devices,
        mqtt_sent,
        mqtt_received,
        cloud_received,
        round(df["Sensor Value"].mean(), 2),
        round(df["Sensor Value"].max(), 2),
        round(df["Sensor Value"].min(), 2)
    ]
})

# 1. Sensor data
print("\n1. PERCEPTION LAYER: DEVICE/SENSOR DATA")
print("-" * 75)
```

```
print(df.to_string(index=False))

# 2. Four-layer flow
print("\n2. FOUR-LAYER IoE DATA FLOW")
print("-" * 75)
print(layer_df.to_string(index=False))

# 3. MQTT output
print("\n3. NETWORK AND MQTT MESSAGING")
print("-" * 75)
print("MQTT messages sent     :", mqtt_sent)
print("MQTT messages received :", mqtt_received)

# Illustrative MQTT payload
if len(df) > 0:
    sample = df.iloc[0]

    print("\nExample MQTT topic:")
    print("ioe/sensors/device01")

    print("\nExample MQTT payload:")
    print({
        "device_id": sample["Device ID"],
        "temperature": sample["Sensor Value"],
        "unit": sample["Unit"]
    })

# 4. Dashboard
print("\n4. APPLICATION LAYER: CLOUD DASHBOARD")
print("-" * 75)
print(dashboard.to_string(index=False))

# Graph 1: Sensor readings
plt.figure(figsize=(9, 4))
plt.plot(df["Device ID"], df["Sensor Value"], marker="o")
plt.title("IoE Sensor Data Monitoring")
plt.xlabel("Device ID")
plt.ylabel("Temperature (°C)")
plt.xticks(rotation=45)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

# Graph 2: Messages through the four layers
```

```
plt.figure(figsize=(9, 5))
x = np.arange(len(layer_df))
width = 0.35

plt.bar(
    x - width/2,
    layer_df["Input Messages"],
    width,
    label="Input Messages"
)

plt.bar(
    x + width/2,
    layer_df["Processed Messages"],
    width,
    label="Processed Messages"
)

plt.xticks(x, layer_df["IoE Layer"])
plt.xlabel("IoE Architecture Layer")
plt.ylabel("Number of Messages")
plt.title("Message Flow Through Four IoE Layers")
plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

# Graph 3: Configured layer success rates
plt.figure(figsize=(8, 4))
plt.bar(
    layer_df["IoE Layer"],
    layer_df["Configured Success Rate (%)"]
)
plt.title("Configured Processing Success by Layer")
plt.xlabel("IoE Layer")
plt.ylabel("Success Rate (%)")
plt.ylim(0, 105)
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

# Graph 4: MQTT and cloud message counts
plt.figure(figsize=(8, 4))
plt.bar(
```

```
    ["Generated", "MQTT Sent", "MQTT Received", "Cloud Received"],
    [
        number_of_devices,
        mqtt_sent,
        mqtt_received,
        cloud_received
    ]
)
plt.title("IoE Message Delivery to the Cloud")
plt.xlabel("Integration Stage")
plt.ylabel("Number of Messages")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

# Graph 5: Dashboard summary
plt.figure(figsize=(8, 4))
plt.bar(
    ["Average", "Maximum", "Minimum"],
    [
        df["Sensor Value"].mean(),
        df["Sensor Value"].max(),
        df["Sensor Value"].min()
    ]
)
plt.title("Cloud Dashboard: Temperature Summary")
plt.xlabel("Temperature Statistic")
plt.ylabel("Temperature (°C)")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

print("\nCONCLUSION")
print("-" * 75)
print("Perception: Generates sensor/device data.")
print("Network: Models communication and MQTT messaging.")
print("Middleware: Models data processing and integration.")
print("Application: Displays cloud dashboard metrics.")
print("Experiment B completed.")
```

OUTPUTS:

```
================================================================================
EXPERIMENT B: FOUR-LAYER IoE ARCHITECTURE SIMULATION
================================================================================

1. PERCEPTION LAYER: DEVICE/SENSOR DATA
--------------------------------------------------------------------------------
Device ID  Sensor Value Unit
DEVICE_01          27.49   °C
DEVICE_02          39.01   °C
DEVICE_03          34.64   °C
DEVICE_04          31.97   °C
DEVICE_05          23.12   °C
DEVICE_06          23.12   °C
DEVICE_07          21.16   °C
DEVICE_08          37.32   °C
DEVICE_09          32.02   °C
DEVICE_10          34.16   °C
DEVICE_11          20.41   °C
DEVICE_12          39.40   °C
DEVICE_13          36.65   °C
DEVICE_14          24.25   °C
DEVICE_15          23.64   °C
DEVICE_16          23.67   °C
DEVICE_17          26.08   °C
DEVICE_18          30.50   °C
DEVICE_19          28.64   °C
DEVICE_20          25.82   °C
```

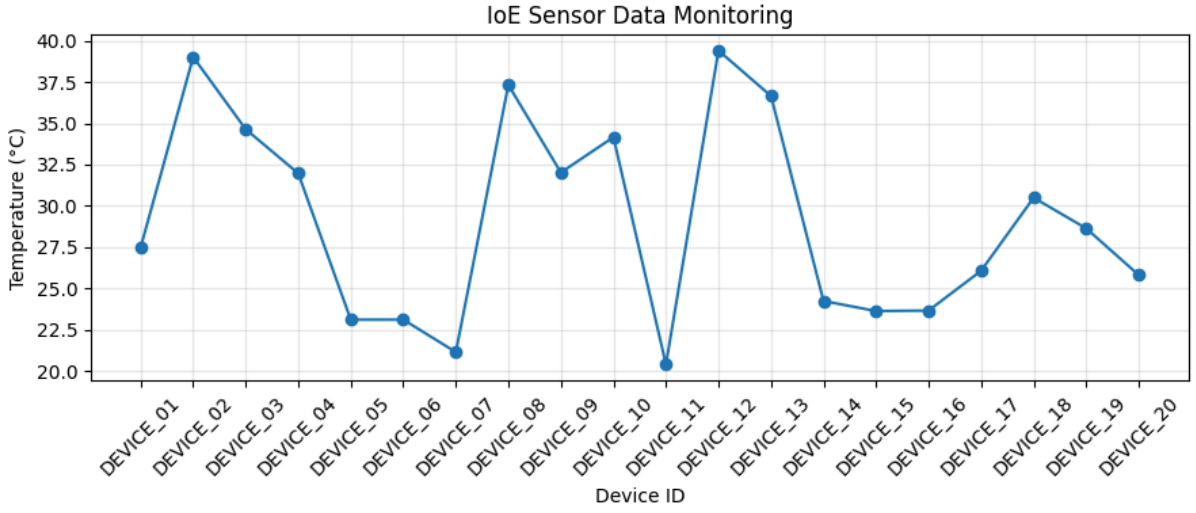
```
2. FOUR-LAYER IoE DATA FLOW
--------------------------------------------------------------------------------
  IoE Layer  Input Messages  Processed Messages  Configured Success Rate (%)
 Perception              20                  20                           98
    Network              20                  19                           97
 Middleware              19                  19                           99
Application              19                  19                           98

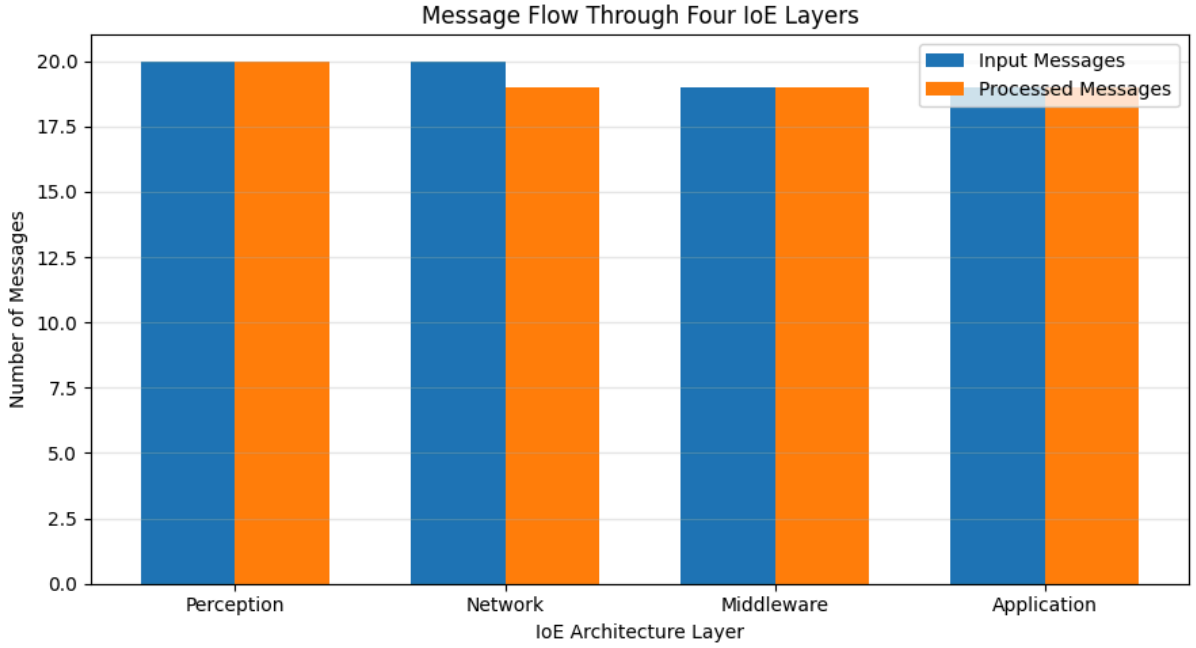
3. NETWORK AND MQTT MESSAGING
--------------------------------------------------------------------------------
MQTT messages sent     : 19
MQTT messages received : 18

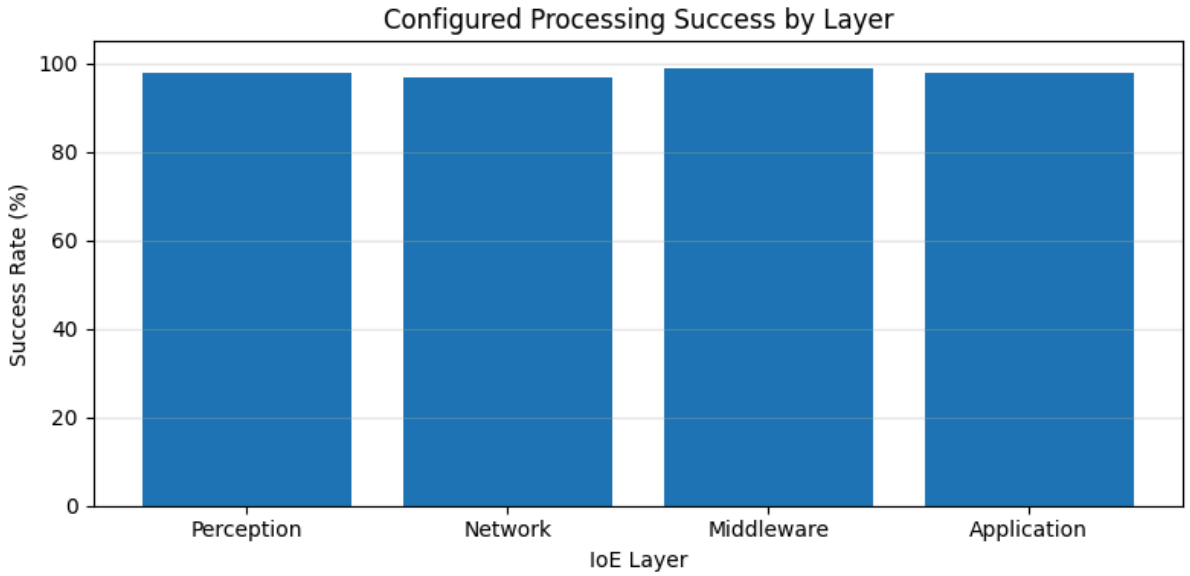
Example MQTT topic:
ioe/sensors/device01

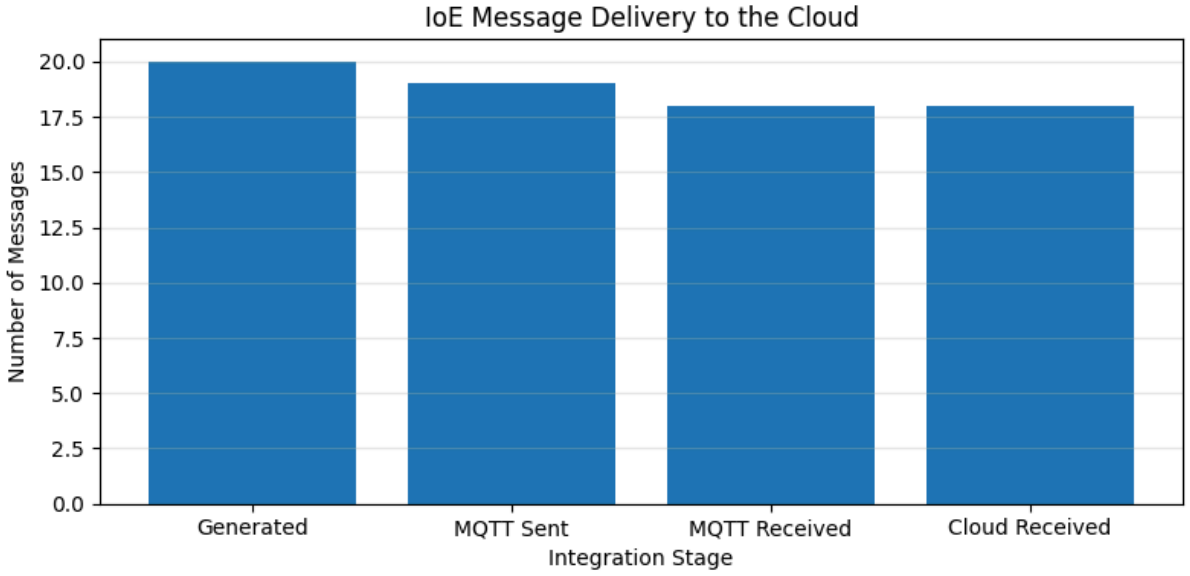
Example MQTT payload:
{'device_id': 'DEVICE_01', 'temperature': np.float64(27.49), 'unit': '°C'}
```

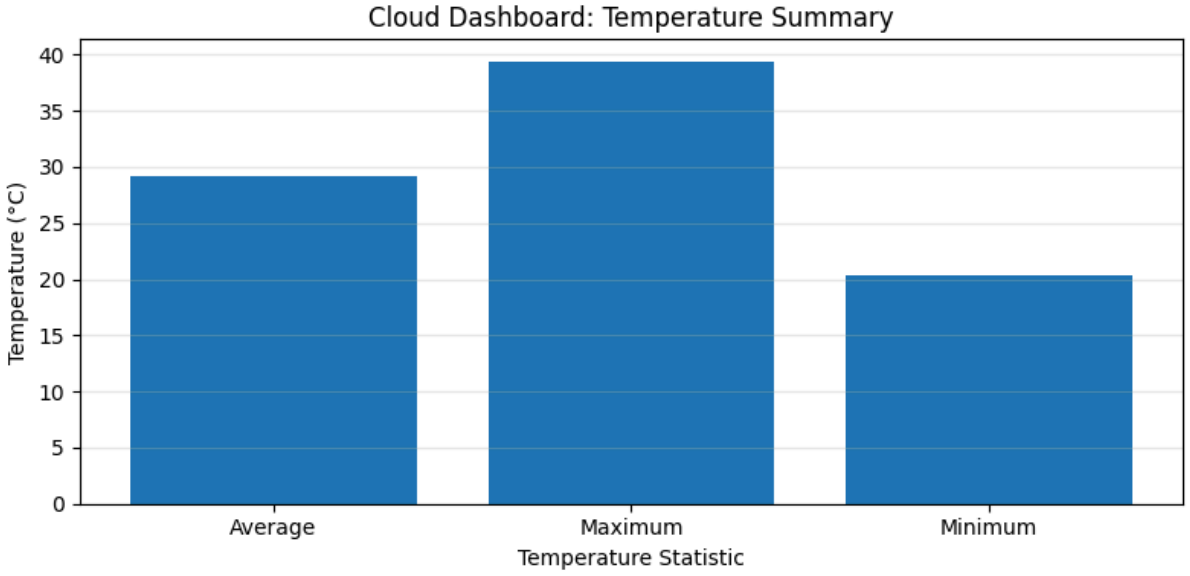
```
4. APPLICATION LAYER: CLOUD DASHBOARD
--------------------------------------------------------------------------------
         Dashboard Metric  Value
        Connected Devices  20.00
Sensor Messages Generated  20.00
       MQTT Messages Sent  19.00
   MQTT Messages Received  18.00
   Cloud Records Received  18.00
Average Sensor Value (°C)  29.15
Maximum Sensor Value (°C)  39.40
Minimum Sensor Value (°C)  20.41
```






Configured Processing Success by Layer
Success Rate (%)
0
20
40
60
80
100
Perception
Network
Middleware
Application
IoE Layer


IoE Message Delivery to the Cloud
Number of Messages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Generated
MQTT Sent
MQTT Received
Cloud Received
Integration Stage


Cloud Dashboard: Temperature Summary
Temperature (°C)
0
5
10
15
20
25
30
35
40
Average
Maximum
Minimum
Temperature Statistic

CONCLUSION

Perception: Generates sensor/device data.
Network: Models communication and MQTT messaging.
Middleware: Models data processing and integration.
Application: Displays cloud dashboard metrics.
Experiment B completed.